Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Simplification of algebraic expressions: order of operations, brackets and word problems. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Second Term
Week	9
Topic	Simplification of Algebraic Terms
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems
Theme	—
Curriculum objectives	• carry out order of operation in algebra • insert/remove bracket and simplify expression • solve word problems involving algebraic expression
Previous knowledge	• Addition and subtraction of algebraic terms • Multiplication and division of algebraic terms
Lesson sequence	• carry out order of operation in algebra — the teacher solves some questions in order of operation on the board; the student listen, contribute, and where necessary ask questions where they do not understand; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$ • insert/remove bracket and simplify expression — the teacher solves some questions on insertion, removal of bracket on the board; the student listen, ask questions on their areas of difficulty; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, marks their notes and makes corrections. Set for the students: (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$ • solve word problems involving algebraic expression — the teacher solves some questions on word problems involving algebraic expression on the board; the student listen, ask questions where they do not understand; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.
Instructional materials	• Chart showing algebraic expression.
Evaluation	• Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$ • Simplify $7 \times 3a - (3a + 5a) \times 2$

	• Simplify $12-(7-2f)-(3-f)$ • Simplify $(8a-5)+(14-3a)$ • A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

Lesson objectives

1.	carry out order of operation in algebra
2.	insert/remove bracket and simplify expression
3.	solve word problems involving algebraic expression

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Simplification of algebraic expressions: order of operations, brackets and word problems

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: Order of Operations in Algebra

When you simplify an algebraic expression, you cannot just work from left to right. You must follow the order of operations, which tells you which part to work out first.

The correct order is:

1. **Brackets** — work out anything inside them first.
2. **Multiplication and division** — do these from left to right.
3. **Addition and subtraction** — do these from left to right.

You already know how to add, subtract, multiply and divide algebraic terms. Order of operations tells you the order in which to apply those skills. In an expression like $3x \times 2 + 5x$, you must multiply before you add, so you deal with $3x \times 2$ first and only then combine the x terms.

Example 1. Find the value of each expression by following the order of operations.

(a) $18 - 6 \times 2$

Multiplication comes before subtraction, so work out 6×2 first.

$18 - 6 \times 2 = 18 - 12 = 6$

(b) $6 - 18 \div 3$

Division comes before subtraction, so work out $18 \div 3$ first.

$6 - 18 \div 3 = 6 - 6 = 0$

(c) $6 \times 2 - 2 - 40 \div 4$

Deal with the multiplication and division first, then subtract from left to right.

$6 \times 2 - 2 - 40 \div 4 = 12 - 2 - 10 = 10 - 10 = 0$

Example 2. Simplify the following expressions.

(a) $3x \times 2 + 5x$

Multiply first: $3x \times 2 = 6x$. Then add the x terms.

$3x \times 2 + 5x = 6x + 5x = 11x$

(b) $17y - 5 \times 2y$

Multiply first: $5 \times 2y = 10y$. Then subtract.

$17y - 5 \times 2y = 17y - 10y = 7y$

Example 3. Simplify the following expressions.

(a) $7x \div x + 5$

Division comes before addition. $\frac{7x}{x} = 7$, then add 5.

$7x \div x + 5 = 7 + 5 = 12$

(b) $4x - (6 \div 3)$

Work out the bracket first: $6 \div 3 = 2$. Then subtract.

$4x - (6 \div 3) = 4x - 2$

(c) $(24x \div 6) + x + (1 \times 5x)$

Work out each bracket first: $24x \div 6 = 4x$ and $1 \times 5x = 5x$. Then add all the x terms.

$(24x \div 6) + x + (1 \times 5x) = 4x + x + 5x = 5x + 5x = 10x$

Practice

1. Simplify: $4n + 3n \times 10$
2. Simplify: $32 + 8x \div 2$
3. Simplify: $3 \times 5x + 4x \div 2$
4. Simplify: $36y \div 6 + y + 2 \times 4y$

Section 2: Removing Brackets to Simplify Expressions

When an expression contains brackets, you can remove them and then collect the like terms. Three hints will guide you every time:

- Always try to simplify the terms inside the bracket first.
- A + sign just before a bracket does **not** change the signs of the terms inside the bracket when the bracket is removed.
- A − sign just before a bracket **changes every sign** inside the bracket when the bracket is removed: each + becomes − and each − becomes +.

Here is what that looks like in the simplest case. In $a + (b + c)$, the plus before the bracket leaves the signs alone, so $a + (b + c) = a + b + c$. In $a + (b - c)$ the same rule applies: $a + (b - c) = a + b - c$.

But in $2a - (3b - c)$, the minus before the bracket flips both signs inside: $3b$ keeps its term but its sign becomes $-3b$, and $-c$ becomes $+c$. So $2a - (3b - c) = 2a - 3b + c$.

Simplify the following.

(a) $6x + (3 - x)$

The plus before the bracket leaves the signs alone, so
$6x + (3 - x) = 6x + 3 - x$.
Collect the like terms: $6x$ and $-x$ are both in x, so
$6x - x + 3 = 5x + 3$.

(b) $(3a + 2) + (5a + a)$

Remove the brackets without changing any signs:
$3a + 2 + 5a + a$.
Collect the like terms: $3a + 5a + a = 9a$, so the answer is
$9a + 2$.

(c) $10d - (9c - 4d)$

The minus before the bracket flips both signs inside: $-9c$ and $+4d$.
So $10d - 9c + 4d$.
Collect the like terms: $10d + 4d = 14d$, so
$14d - 9c$.

(d) $(6x - y) - (4x - 5y)$

Remove the first bracket as it is. The second bracket is preceded by a minus, so $4x$ becomes $-4x$ and $-5y$ becomes $+5y$:
$6x - y - 4x + 5y$.
Collect the like terms: $6x - 4x = 2x$ and $-y + 5y = 4y$, so
$2x + 4y$.

Practice

1. Simplify $(12a - 7) + (4 - 5a)$.
2. Simplify $(8x - 2y) - (9x - 4y)$.
3. Simplify $12 - (7 - 2f) - (3 - f)$.
4. Simplify $(8a - 5) + (14 - 3a)$.

Section 3: Word Problems with Algebraic Expressions

A word problem tells a short story, and your job is to turn that story into an algebraic expression. You do this by deciding which part of the story is your unknown (the letter), then writing the other costs or numbers in terms of it. You already know how to add, subtract, multiply and divide algebraic terms, and you know how to remove brackets and simplify. A word problem just asks you to use those skills on a story.

Here is the plan to follow for any word problem:

1. Read the story and decide which item the letter stands for.
2. Write that item's value using the letter.
3. Write every other item as the letter, plus or minus a number, depending on the words "more than" or "less than".
4. Add the items together when the problem asks for a total.
5. Simplify by collecting like terms and removing brackets.

Remember the language of the problems: "more than" means add, and "less than" means subtract. If a pencil costs 15 naira less than a pen, then the pencil is the pen's cost minus 15.

Let's work through a problem together.

A pen costs ₦P and a pencil costs 15 naira less than a pen.

(a) Find the cost of a pencil.
(b) Find the cost of a pen and a pencil.

We let the pen be the letter, so the cost of a pen is ₦P.

(a) The pencil costs 15 naira less than the pen, so
cost of a pencil = ₦(P − 15).

(b) The cost of a pen and a pencil is the two costs added together:
cost = ₦P + ₦(P − 15)
= ₦P + ₦P − ₦15
= ₦2P − ₦15
= ₦(2P − 15).

So the pencil costs ₦(P − 15), and the pen together with the pencil costs ₦(2P − 15).

Here is a second example, this time with "more than".

A pencil costs ₦x and a rubber costs 40 naira more than a pencil.

(a) Find the cost of a rubber.
(b) Find the cost of a pencil and a rubber.

We let the pencil be the letter, so the cost of a pencil is ₦x.

(a) The rubber costs 40 naira more than the pencil, so
cost of a rubber = ₦(x + 40).

(b) The cost of a pencil and a rubber is the two costs added:
cost = ₦x + ₦(x + 40)
= ₦(x + x + 40)
= ₦(2x + 40).

So the rubber costs ₦(x + 40), and the pencil together with the rubber costs ₦(2x + 40).

Word problems are not only about money — the letter can stand for a number of people too.

In a class there are x boys and y girls last semester. This semester, 5 boys join and 3 girls leave the class. How many students are now in this class?

The total number of students is the number of boys plus the number of girls.

Number of boys now = x + 5 (5 boys joined).
Number of girls now = y − 3 (3 girls left).

Total number = x + 5 + y − 3
= x + y + 5 − 3
= x + y + 2.

So there are now x + y + 2 students in the class.

Practice

1. A pen costs h naira and a crayon costs ₦20 less than a pen.
   (a) Find the cost of a crayon.
   (b) Find the cost of a crayon and a pen.
2. Envelopes cost d naira per pack, and 4 notepads together cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelopes and one notepad.
3. A pen and a pencil altogether cost x naira. If a pen costs y naira, find the cost of 3 pencils.

Check your work

Attempt every question before you read this page.

Order of Operations in Algebra

1. $34n$

   Multiply first: $3n \times 10 = 30n$. Then add: $4n + 30n = 34n$.
2. $32 + 4x$

   Divide first: $8x \div 2 = 4x$. Then add: $32 + 4x$. There are no x terms to combine with 32, so the answer is $32 + 4x$.
3. $17x$

   Multiply first: $3 \times 5x = 15x$. Divide next: $4x \div 2 = 2x$. Then add: $15x + 2x = 17x$.
4. $15y$

   Work out the division and multiplication first: $36y \div 6 = 6y$ and $2 \times 4y = 8y$. Then add all the y terms from left to right: $6y + y + 8y = 7y + 8y = 15y$.

Removing Brackets to Simplify Expressions

1. $7a - 3$

   No signs change when the brackets are removed:
   $12a - 7 + 4 - 5a$.
   Collect the like terms: $12a - 5a = 7a$ and $-7 + 4 = -3$.
   So the answer is $7a - 3$.
2. $-x + 2y$

   Remove the first bracket as it is. The second bracket is preceded by a minus, so $9x$ becomes $-9x$ and $-4y$ becomes $+4y$:
   $8x - 2y - 9x + 4y$.
   Collect the like terms: $8x - 9x = -x$ and $-2y + 4y = 2y$.
   So the answer is $-x + 2y$.
3. $2 + 3f$

   Remove the first bracket: the minus before it flips 7 to -7 and $-2f$ to $+2f$. Remove the second bracket: the minus flips 3 to -3 and $-f$ to $+f$.
   So we get $12 - 7 + 2f - 3 + f$.
   Collect the number terms: $12 - 7 - 3 = 2$. Collect the f terms: $2f + f = 3f$.
   So the answer is $2 + 3f$.
4. $5a + 9$

   No signs change when the brackets are removed:
   $8a - 5 + 14 - 3a$.
   Collect the like terms: $8a - 3a = 5a$ and $-5 + 14 = 9$.
   So the answer is $5a + 9$.

Word Problems with Algebraic Expressions

1. (a) cost of a crayon = ₦(h − 20). (b) cost of a crayon and a pen = ₦(2h − 20).

The pen is the letter, so the pen costs ₦h. (a) The crayon costs 20 less than the pen, so it costs ₦(h − 20). (b) Add the two: ₦h + ₦(h − 20) = ₦h + ₦h − ₦20 = ₦(2h − 20).

2. ₦(5d + 20)/4

   A pack of envelopes costs ₦d. The 4 notepads together cost 20 naira more, so 4 notepads cost ₦(d + 20), which means one notepad costs ₦(d + 20)/4. Total for a pack of envelopes and one notepad = ₦d + ₦(d + 20)/4 = ₦(4d)/4 + ₦(d + 20)/4 = ₦(4d + d + 20)/4 = ₦(5d + 20)/4.

3. ₦3(x − y)

   The pen and pencil together cost x naira, and the pen costs y naira, so one pencil costs ₦(x − y). Three pencils therefore cost ₦3(x − y).

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-d102fed7	**Date**	
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____